21CSC203P– ADVANCED PROGRAMMING PRACTICE

[PROJECT TITLE IN CAPITAL LETTERS]

A PROJECT REPORT

Submitted by

[STUDENT NAME 1] [Reg. No: RA0000000000000]

[STUDENT NAME 2] [Reg. No: RA0000000000000]

[STUDENT NAME 3] [Reg. No: RA0000000000000]

[STUDENT NAME 4] [Reg. No: RA0000000000000]

[Add or delete lines according to team size]

Under the guidance of

Dr. [GUIDE NAME]

(Assistant Professor, Department of Computer Science and Engineering)

In partial fulfilment of the requirements for the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

SCHOOL OF COMPUTING

FACULTY OF ENGINEERING AND TECHNOLOGY

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

TIRUCHIRAPPALLI – 621 105

NOVEMBER 2026

DECLARATION

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

> *Instruction: for an individual project, change "We / our" to "I / my". Delete this line.*

Name	Register Number	Signature
[Student Name 1]	RA0000000000000	
[Student Name 2]	RA0000000000000	
[Student Name 3]	RA0000000000000	
[Student Name 4]	RA0000000000000	

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

TIRUCHIRAPPALLI

(Deemed to be University u/s 3 of UGC Act, 1956)

BONAFIDE CERTIFICATE

Certified that this project report titled "[PROJECT TITLE]" is the bonafide work of "[STUDENT NAME 1] [Reg. No.], [STUDENT NAME 2] [Reg. No.], [STUDENT NAME 3] [Reg. No.], [STUDENT NAME 4] [Reg. No.]" who carried out the project work under my supervision. Certified further that, to the best of my knowledge, the work reported herein does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

SIGNATURE

Dr. [GUIDE NAME]
GUIDE
Assistant Professor
Dept. of CSE

SIGNATURE

Dr. S. Kanaga Suba Raja
HEAD OF THE DEPARTMENT
Dept. of CSE

Signature of the Examiner

Date: Place: Tiruchirappalli

ABSTRACT

[Type your abstract here. The abstract is a one-page synopsis of the project report. State the problem and its importance, the approach or system developed, the tools and techniques used, and the main results with key figures (accuracy, response time, etc.) in about 200 to 250 words. Do not include citations, figures or tables in the abstract.]

Keywords: [keyword 1; keyword 2; keyword 3; keyword 4]

> *Instruction: the abstract must fit on one page (Times New Roman 14 pt, double line spacing). Delete this line.*

ACKNOWLEDGEMENTS

We express our sincere gratitude to our guide, Dr. [GUIDE NAME], Assistant Professor, Department of Computer Science and Engineering, for the valuable guidance, constant encouragement and timely support throughout this project.

We thank Dr. S. Kanaga Suba Raja, Head of the Department, for providing the facilities and the opportunity to carry out this work. We also thank the faculty members of the department, our friends and our families for their help and encouragement.

> *Instruction: keep the acknowledgement to one page. Edit the wording to suit your team. Delete this line.*

[Student names]

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ABBREVIATIONS

> *Instruction: list every abbreviation used in the report in alphabetical order. Delete the example rows.*

Abbreviation	Full form
[ABBR]	[Full form]
[ABBR]	[Full form]
[ABBR]	[Full form]

LIST OF SYMBOLS

Optional. Use this page only if the report contains mathematical symbols. List them in alphabetical order with units. Delete this page otherwise.

Symbol	Meaning, unit
[symbol]	[Meaning, unit]
[symbol]	[Meaning, unit]

1 INTRODUCTION

1.1 Background and Introduction

> *Instruction: introduce the domain and the technology behind the project in 2 to 3 paragraphs. Explain why the topic matters. Cite sources as [1], [2].*

1.2 Problem Statement

> *Instruction: state the problem in a short paragraph. Say what is wrong or missing today and who is affected.*

1.3 Objectives

> *Instruction: list 4 to 8 specific, measurable objectives, each starting with a verb (design, implement, evaluate ...).*

- [Objective 1]
- [Objective 2]
- [Objective 3]

1.4 Scope and Expected Outcome

> *Instruction: describe what is covered and what is not (limits of the project), and the final outcome the project delivers.*

1.5 Organization of the Report

> *Instruction: one short paragraph describing what each chapter contains.*

2 LITERATURE SURVEY

2.1 Review of Existing Work

> *Instruction: review at least 8 to 10 recent papers, books or technical documents related to your project. Group them by approach or theme. Paraphrase in your own words and cite each source.*

2.2 Summary of Related Work

> *Instruction: summarise the reviewed work in a table (Table 1 below is an example). Delete the example rows.*

Table 1. Summary of related work

Ref.	Author(s), Year	Method / Technique	Limitation
[1]	[Author, Year]	[Method]	[Limitation]
[2]	[Author, Year]	[Method]	[Limitation]
[3]	[Author, Year]	[Method]	[Limitation]

2.3 Research Gap

> *Instruction: state clearly the gap or limitation in existing work that your project addresses.*

3 SYSTEM ANALYSIS

3.1 Existing System and its Limitations

> *Instruction: describe how the problem is handled today and where it falls short.*

3.2 Proposed System

> *Instruction: describe your solution and how it overcomes the limitations above. List its main advantages.*

3.3 Requirement Analysis

Functional Requirements

> *Instruction: list what the system must do.*

Non-Functional Requirements

> *Instruction: list performance, security, usability, reliability and similar requirements.*

3.4 Feasibility Study (optional)

> *Instruction: briefly discuss technical, economic and operational feasibility. Delete this section if not applicable.*

3.5 Hardware and Software Requirements

> *Instruction: list the hardware, software and tools used (Table 2 below is an example). Delete the example rows.*

Table 2. Hardware and software requirements

Category	Component	Purpose
Hardware	[Processor, RAM, storage]	[Purpose]
Operating System	[Name and version]	[Purpose]
Programming Language	[Name and version]	[Purpose]
Tools / Libraries	[Names]	[Purpose]

4 SYSTEM DESIGN

4.1 System Architecture

> *Instruction: present the overall architecture diagram and explain each component and how they interact. Refer to the figure in the text (for example, "Figure 1 shows ...").*

[Insert architecture diagram here. Centre it; keep the width within the margins.]

Figure 1. System architecture of the proposed system

4.2 Modules and Components

> *Instruction: describe each module or component: its inputs, processing and outputs.*

4.3 Workflow and Design Diagrams

> *Instruction: include the diagrams relevant to your project: flowchart, data flow diagram, use case, class, sequence, ER diagram, network topology, circuit diagram. Explain each one.*

4.4 Dataset / Database Design (optional)

> *Instruction: for data-driven projects, describe the dataset (source, size, classes, pre-processing) or the database schema. Delete this section if not applicable.*

5 IMPLEMENTATION

5.1 Development Environment and Tools

> *Instruction: describe how the environment was set up (installation, configuration, versions). Add screenshots where helpful.*

5.2 Module-wise Implementation

> *Instruction: explain how each module from Chapter 4 was implemented, step by step.*

5.3 Key Algorithms and Code Snippets

> *Instruction: present the main algorithm or logic, with only the important code snippets (about 10 to 20 lines each). Put the full source code in Appendix A.*

```
# Example: paste a short, important code snippet here
def main():
    pass
```

5.4 Output Screenshots

> *Instruction: show the working output with screenshots. Every screenshot needs a caption and must be explained in the text.*

6 TESTING

6.1 Test Environment and Setup

> *Instruction: describe the environment, data and tools used for testing.*

6.2 Test Cases and Scenarios

> *Instruction: list the test cases (Table 3 below is an example). Cover normal operation, boundary conditions and failure or recovery scenarios. Delete the example rows.*

Table 3. Test cases and results

TC ID	Description	Input	Expected Output	Actual Output	Status
TC01	[Description]	[Input]	[Expected]	[Actual]	Pass
TC02	[Description]	[Input]	[Expected]	[Actual]	Pass

6.3 Test Results

> *Instruction: summarise the outcome of testing, any defects found and how they were fixed. Include evidence (logs, screenshots).*

7 RESULTS AND DISCUSSION

7.1 Experimental Results

> *Instruction: present the measured results as tables and graphs (Table 4 and Figure 2 below are examples). State the number of runs or samples, and the conditions of the experiment.*

Table 4. Experimental results

Parameter	Value	Remarks
[Metric 1]	[Value]	[Remark]
[Metric 2]	[Value]	[Remark]

[Insert graph or chart here. Label both axes with units and add a legend if there is more than one series.]

Figure 2. Performance of the proposed system

7.2 Performance Analysis

> *Instruction: interpret the results. Explain trends, unexpected values and the factors that affected them.*

7.3 Comparative Analysis

> *Instruction: compare your results with the existing systems or the papers reviewed in Chapter 2, preferably in a table.*

7.4 Discussion and Limitations

> *Instruction: discuss what the results mean, how far the objectives were met, and the limitations of the work.*

8 CONCLUSION AND FUTURE ENHANCEMENT

8.1 Conclusion

> *Instruction: summarise what was done and achieved, with key numbers, and relate it back to the objectives in Chapter 1. Do not introduce new material.*

8.2 Future Enhancement

> *Instruction: list realistic improvements and extensions of the work.*

- [Enhancement 1]
- [Enhancement 2]

REFERENCES

> *Instruction: use IEEE style, numbered in the order of first citation. Replace the three examples below with your own sources and delete this line.*

[1] [A. Author, B. Author and C. Author, “Title of the paper,” Journal Name, vol. 10, no. 2, pp. 100–110, Year.] (journal article)

[2] [A. Author, Title of the Book, Edition. City, Country: Publisher, Year.] (book)

[3] [Organization or Author, “Title of the web page,” Website Name. URL (accessed Month Day, Year).] (website or documentation)

APPENDIX A: SOURCE CODE

> *Instruction: include the complete source code (or the main files) in Courier New 10 pt. Mention the file names. If the code is very long, include only the main modules and give the repository link.*

Repository link: [GitHub / demo link, if any]

```
# [file name]
# paste source code here
```

APPENDIX B: ADDITIONAL EVIDENCE

> *Instruction: add extra screenshots, logs, datasheets, user manual or sample forms that support the report but are too long for the main chapters.*

APPENDIX C: SIMILARITY (PLAGIARISM) REPORT

> *Optional. Attach the similarity report here only if the department requires it. Delete this appendix otherwise.*